भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 352: Computer Networks, Mid-Semester Examination, 23 September 2024

The question paper has two parts : Part A and Part B.

Instructions for Part A: Maximum time: 1 hour. Maximum marks: 50.

The answer to each question must be given in a separate answer sheet given to you, attached to the question paper. Do not ask the instructor any questions regarding the paper. Make reasonable assumptions where necessary and state them. State mistakes in the question paper, if any, in your answer sheet. 2 marks will be awarded for a correct answer and -0.5 for an incorrect answer. If a question is skipped, 0 marks will be awarded. For MCQs, there is only one correct . Submit the answer sheet for this part within the first hour of the exam.

Instructions for Part B: Maximum time: 1 hour. Maximum marks: 40.

Answers must be brief and to the point. Marks will be awarded only if the necessary justifications are provided, wherever they are sought. For numerical questions, marks will be awarded only if the steps are shown, whether the final answer is correct or not. Do not ask the instructor any questions regarding the paper. Make reasonable assumptions where necessary and state them. State mistakes in the question paper, if any, in your answer sheet.

1 Part A

1. Consider the following multiple access protocols that were taught in class: CSMA, Aloha, and Polling. Which of these protocols are collision-free (that is, collisions will never happen)? A. Aloha B. CSMA/CA C. CSMA/CD **D. Polling**

2. Choose the best reason below for using link-layer ACKs in 802.11 networks A. 802.11 uses omni-directional antennas. **B. Because of the hidden terminal problem, a node that is transmitting and hears no collisions still doesn't know if there was a collision at the receiver.** C. All other stations will stop transmitting when they hear an ACK. This reduces collisions. D. The sender will measure the signal strength of the received ACK and will decide if the receiver is moving towards or away from it.

3. Suppose an Ethernet frame arrives at a layer-2 switch. The switch does not know which of its ports leads to the node with the destination MAC address in the frame. What does the switch do? A. Send an ARP broadcast message to determine the appropriate port **B. Send the frame to all the ports except the one on which it was received** C. Drop the frame D. Forward the frame to the destination port given in the frame

4. Consider the LAN in Fig. 1 with a learning switch. Assume that the switch table is initially empty. A frame is sent from host C to host C'. Then C' replies back to C. How many of these two frames are also received at the interface to B? A. 0 **B. 1** C. 2 D. 3

5. Which of the following is FALSE with regard to the network layer? A. It is implemented in routers B. It is implemented in hosts at the network edge **C. The network layer**

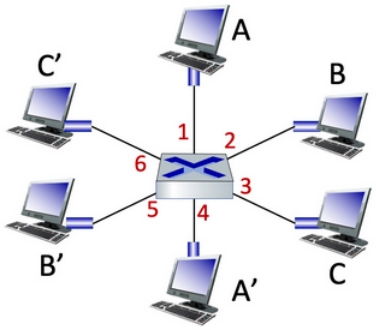


Figure 1: Star topology

is implemented in Ethernet switches in a local area network. D. The network layer may fragment packets

6. Which of the following quality-of-service guarantees is part of the Internet's best-effort service model? A. Guaranteed delivery from sending host to receiving host. B. A guaranteed minimum bandwidth is provided to a source-to-destination flow of packets C. In-order datagram payload delivery to the transport layer of those datagrams arriving at the receiving host **D. None of the above**

7. Which of the following is true for both pure Aloha and CSMA/CD? A. Pure Aloha and CSMA/CD can achieve 100% channel utilization, in the case that all nodes always have frames to send. **B. Pure Aloha and CSMA/CD can achieve 100% utilization, in the case that there is only one node that always has frames to send** C. A channel is always sensed before transmission D. Collisions are always avoided

8. Match the following:

   1. Time needed to perform an integrity check, lookup packet information in a local table and move the packet from an input link to an output link in a router. 2. Time spent waiting in packet buffers for link transmission. 3. Time spent transmitting packets bits into the link. 4. Time need for bits to physically propagate through the transmission medium from end one of a link to the other.

   P. Queueing delay Q. Processing delay R. Transmission delay S. Propagation delay

   A. 1-P, 2-Q, 3-S, 4-R **B. 1-Q, 2-P, 3-R, 4-S** C. 1-R, 2-Q, 3-P, 4-S D. None of the above

9. Attenuation distortion is not much of a problem with digital signals. This is because: A. A digital signal consists only of low frequency components. B. The low frequency components of a digital signal are low in strength C. All constituent frequencies of a digital signal travel at the same speed. **D. The high frequency components of a digital signal are low in strength.**

10. We need to upgrade a channel to a higher capacity. Suppose we double the bandwidth, Choose all that apply. **A. The data rate is doubled** B. The new data rate is 4 times the old data rate C. The data rate is halved. D. The new data rate is 0.3 times the old data rate

11. Two hosts use the Stop-and-Wait ARQ protocol. They are connected by a point-to-point link. Under which of these circumstances is the utilization of the link the lowest? A. Shorter link length and higher transmission rate B. Shorter link length and lower transmission rate C. Longer link length and lower transmission rate **D. Longer link length and higher transmission rate**

12. Assume that two hosts communicate using the Go-Back-N ARQ protocol. The protocol runs over a best-effort network, but no packet or ACK is duplicated at the network or link layers. The sender sends a stream of new packets according to the sliding window protocol, and in response gets the following cumulative ACKs from the receiver:1 2 3 4 4 4 4 4 4 4

    Now suppose that the sender times out and retransmits the first unacknowledged packet. When the receiver gets that retransmitted packet, what can you say about the ACK, a, that it sends?Assume that the send window size is greater than 1. Choose the best answer from below.

    A. a = 5. **B.** $a \geq 5$**.** C. $5 \leq a \leq 11$. D. a = 11

13. Which of the following is FALSE with regard to packet switching? A. It is possible to support a connection-oriented service in packet switching B. In a packet-switched network providing a connectionless service, packets carry information in the header that tells the switches about their destination **C. In a connection oriented packet switched network, switches use the destination address information that packets carry** D. In a connectionless network, the router routing tables may change

14. In wireless networks, CSMA/CD is not used. Which of the following is NOT a reason for this? A. Due to signal fading, it is possible that collisions are not detected. B. In a wireless device, it takes a lot of power to receive while transmitting **C. Collisions may not be detected due to the exposed terminal problem** D. None of the above

15. You are tasked with designing a network that uses VLANs to segment traffic for better security and performance. Which of the following statements is NOT true regarding VLANs and their operation? A. VLANs allow multiple logical networks to coexist on a single physical network infrastructure. B. The VLAN tagging protocol adds a header to frames to identify the VLAN to which they belong. **C. All switches in a network must be configured with the same VLANs to ensure proper communication across the network.** D. VLANs can be used to limit broadcast traffic within a network segment.

16. Which of the following is TRUE with regard to carrier extension used in Gigabit Ethernet? **A. It is inefficient when a series of short frames need to be sent.** B. This method

allows decreasing the maximum cable length from a station to a hub compared to the traditional method C. There will never be a need to add extension bits when carrier extension is used D. Extension bits will always need to be added when carrier extension is used

17. The vulnerable time for CSMA is **A. The propagation time between the farthest stations in the network** B. The maximum transmission time of a frame C. Twice the maximum transmission time of a frame D. The notion of vulnerable time is not applicable to CSMA

18. Assume we have a bus network that uses CSMA/CD with only two stations, A and B, in which the frame transmission time is 40μs and the propagation time is 25μs. Station A starts sending a frame at time $t = 0.0\mu$s and station B starts sending a frame at $t = 23.0\mu$s. Which of the following is true? A. The frames collide and both the stations detect the collision B. The frames do not collide C. The frames collide, but only A detects the collision **D. The frames collide, but only B detects the collision**

19. Consider a 100 Mbps link connecting an earth station (sender) to a satellite (receiver) positioned at distance of 2400 km. The signal propagates at a speed of $3 \times 10^8 m/s$. The time required for the receiver to fully receive a packet of 5000 bytes transmitted by the sender, measured in milliseconds and rounded to two decimal places, is _____ **Answer: 8.4**

20. A router whose transmission rate is 1000 Mbps is transmitting packets each of length 16000 bits to another router at the end of the link. The maximum number of packets transmitted per second is _____. **Answer: 62,500**

21. Two hosts A and B are connected by a link of rate 10^6 bits/sec. The distance between the hosts is 10,000 km. The propagatin speed is 2×10^8m/sec. Host A sends a file of 50,000 bytes as one large message to B continuously. The transmission delay is _____ milliseconds. **400**

22. For question 21, the propagation delay is _____ milliseconds. **Answer: 50**

23. Consider a channel with a 1-MHz capacity and an SNR of 63. The upper limit to the data rate that the channel can carry is _____ Mbps. **Answer: 6**

24. The result of question 23 is the upper limit. However, as a practical matter, better error performance will be achieved at a lower data rate. Assume we choose a data rate of 2/3 the maximum theoretical limit. The number of signal levels needed to achieve this data rate is _____ . **Answer: 4**

25. The maximum number of Class A networks possible is _____. **Answer: 128**

2 Part B

1. (a) [2 points] State one advantage and one disadvantage of cut-through switching compared to store-and-forward switching.

   (b) [2 points] Calculate the latency (from first bit sent to last bit received) for a 1-Gbps Ethernet with a single store-and-forward switch in the path and a packet size of 8000 bits. Assume that each link introduces a propagation delay of 10μs and that the switch begins retransmitting immediately after it has finished receiving the packet.

   (c) [2 points] Calculate the latency where the conditions are the same as b but with three switches in the path of the packet.

   (d) [4 points] Calculate the latency where the conditions are the same as c, but assume the switch implements cut-through switching; it is able to begin retransmitting the packet after the first 256 bits have been received.

   Solution: a) 36 μs (2 propagation delays + 2 transmission dealys) b) 72 μs (4 propagation delays + 4 transmission delays) c) If the switches start transmitting the first bit as soon as they receive the first bit, they would have 0 transmission delay. When there is cut-through each of them delays by 256 bits. Therefore the total time is $8\mu s + 0.256\mu s \times 3 + 4 * 10\mu s = 48.768\mu s$.

2. Refer Fig. 2a for the following questions. The transmission range of a station is represented as a circle with the station at the centre of the circle.

   (a) [2 points] When E is transmitting to B, what are the hidden terminals? Justify your answer.

   (b) [2 points] When B is transmitting to D, what are the exposed terminals? Justify your answer.

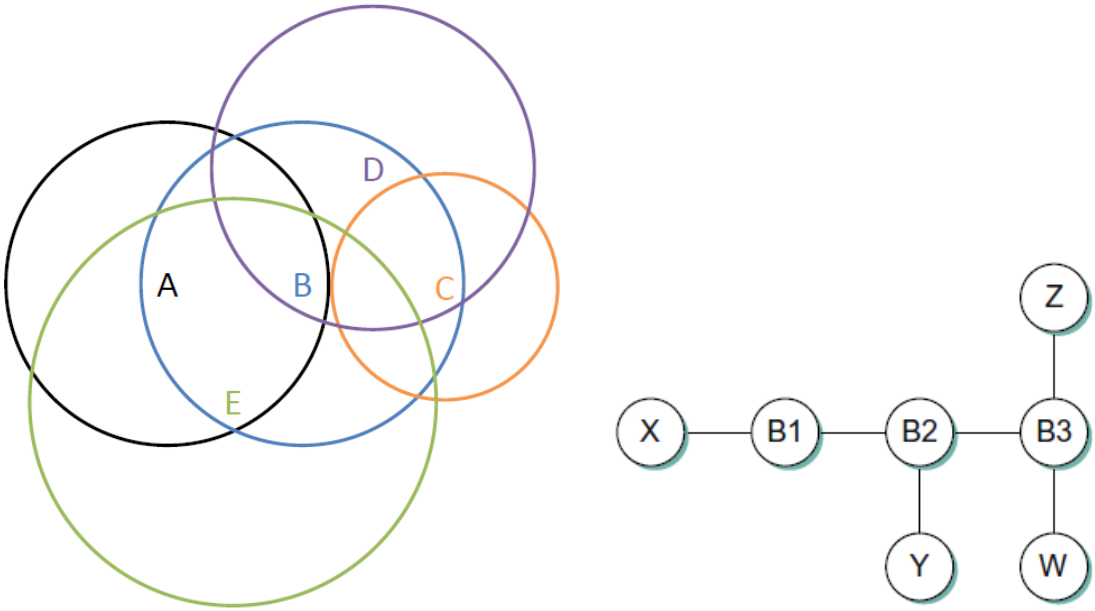
   (c) [4 points] Explain the mechanism in 802.11 to solve the hidden terminal problem.

   Solution: a) D b) A, E and C

3. Refer Fig. 2b for the following questions. Consider hosts X, Y, Z,W and learning bridges B1, B2, B3, with initially empty forwarding tables.

   (a) [2 points] Suppose X sends to W. Which bridges learn where X is? Does Y's network interface see this packet?

   (b) [2 points] Suppose Z now sends to X. Which bridges learn where Z is? Does Y's network interface see this packet?



(a) A wireless network

(b) A layer 2 wired network

Figure 2: Wireless and wired networks

(c) [2 points] Suppose Y now sends to X. Which bridges learn where Y is? Does Z's network interface see this packet?

(d) [4 points] Explain with the help of an example an issue with regard to forwarding packets when learning bridges/switches are used.

Solution: a) B1, B2, B3. Yes. b) B1, B2, B3. No. c) B2, B1. No

4. (a) [2 points] Suppose you are tasked with implementing a Wireless AP. Is it possible at all that you will need to implement both CSMA/CD and CSMA/CA in the Wireless AP? Explain why or why not.

   (b) [4 points] Suppose packets are being sent from host A to host B connected to the same wired LAN which has one layer-2 switch. Will the MAC address of the switch used in the LAN be present in the packets sent on the LAN? Explain which layer 2 addresses will be present in the packets and how they are used.

   (c) [6 points] Suppose A and B are connected to a Wireless AP. Will the MAC address of the AP be present in the packets always? Explain which layer 2 addresses will be present in the packet and under what circumstances. Enumerate all the cases.

Solution: a) Yes, as the Wireless AP may interface with an Ethernet L2 Switch